

Abstract

This paper covers the necessary steps to proving the linear independence between three distinct vector-valued functions by use of proof by contradiction via judicious choice.

Definition 1. Let $f_1, f_2, f_3 : \mathbb{R} \rightarrow \mathbb{R}$ be defined by:

$$f_1(x) = e^x, \quad f_2(x) = e^{-x}, \quad f_3(x) = e^{2x} \quad (1)$$

Lemma 1. *The functions f_1 and f_2 are linearly independent.*

Proof. We proceed by contradiction. If we assume f_1 is dependent on f_2 then:

$$k_1 f_1 + k_2 f_2 = 0 \quad \forall x \in \mathbb{R} \quad (2)$$

We avoid the trivial solution by stating $k_1, k_2 \neq 0$. By judicious choice, we can set up the system evaluated at the points $x \in \{0, 1\}$:

$$\text{At } x = 0 : \quad k_1 + k_2 = 0 \quad (3)$$

$$\text{At } x = 1 : \quad k_1 e + k_2 e^{-1} = 0 \quad (4)$$

This yields: $k_1 = -k_2$. Substituting into Equation (4) gives:

$$-k_2 e + k_2 e^{-1} = 0 \quad (5)$$

$$\implies k_2 (e^{-1} - e) = 0 \quad (6)$$

$$\implies e^{-1} = e \quad (7)$$

This is a contradiction. Dividing both sides yields $(e^{-1} = e)$, which is a false statement. Therefore the set of functions $\{f_1, f_2\}$ are linearly independent. $\square$

Theorem 1. *The set of functions $\{f_1, f_2, f_3\}$ is linearly independent.*

Proof. We proceed by contradiction. Assume the following holds for all $x \in \mathbb{R}$:

$$c_1 f_1(x) + c_2 f_2(x) = f_3(x) \quad (8)$$

By judicious choice, we can evaluate this equation at the points $x \in \{-1, 0, 1\}$ giving us the following system:

$$\text{At } x = 0 : \quad c_1 + c_2 = 1 \quad (9)$$

$$\text{At } x = 1 : \quad c_1 e + c_2 e^{-1} = e^2 \quad (10)$$

$$\text{At } x = -1 : \quad c_1 e^{-1} + c_2 e = e^{-2} \quad (11)$$

From Equation (9), it follows that $c_1 = 1 - c_2$. Finally, evaluating Equation (11) using c_2 results in: Substituting into Equation (10) yields:

$$e - c_2 e + c_2 e^{-1} = e^2 \quad (12) \quad e^{-1} - \frac{e^2 - e}{-e + e^{-1}} e^{-1} + \frac{e^2 - e}{-e + e^{-1}} e = e^{-2} \quad (14)$$

$$\implies c_2 = \frac{e^2 - e}{-e + e^{-1}} \quad (13) \quad \implies \frac{1}{e} + e - e^2 = e^{-2} \quad (15)$$

$$\implies -4.30 \dots = 0.14 \dots \quad (16)$$

This is a contradiction. No real values of c_1, c_2 result in a true statement by Equation (15); therefore the statement in Equation (8) is not true; the set of functions $\{f_1, f_2, f_3\}$ is linearly independent. $\square$